

UN SDG 3 · GOOD HEALTH & WELL-BEING

Grounded AI Health Assistant & Emergency Triage

An auditable retrieval-augmented assistant with rule-based emergency escalation

3

TARGET 3.8

Access to safe, verified health information without medical misinformation.

IT

PROJECT CANDIDATE

**Shardul Shripad
Hingane**

Registration No: **2023bit005**
Department of Information
Technology

PG

PROJECT SUPERVISION

C.P. Navdeti

Project Guide · Dept of IT
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The Critical Problem: Fluent Fabrication in Healthcare AI

1

The Danger of Fluent Fabrication

General LLMs are trained to always generate an answer. A confident, well-structured, completely fabricated paragraph about chest pain is more dangerous than an obvious typo — patients trust it blindly.

2

Total Lack of Provenance

Standard chatbots synthesize knowledge across unvetted internet forums (Reddit, WebMD, blogs). They cannot point to the legal document, paragraph, or clinical body behind a claim.

3

Emergency Neglect

General LLMs engage in conversational chit-chat ('I understand your chest hurts; have you tried resting?'). In acute crises, every minute lost to AI generation can be fatal.

Cost asymmetry: a 5% hallucination rate or a 3-second delay in acute cardiac arrest is catastrophic — code has compiler errors, medicine does not.

System Philosophy: Four Architectural Guardrails

1

Explain, Never Diagnose

The system only explains verified medical literature. It has no capability to diagnose, prescribe medications, or suggest dosages — preventing liability and clinical drift.

2

100% Auditable Provenance

Every factual statement carries an inline citation chip linking directly to a specific passage in a published document from WHO, NHS, NIH, or CDC, with its legal license.

3

Structural Refusal — 0.65 Cosine Floor

If retrieved document similarity is below 0.65, the system refuses to answer. The LLM is never called — this structurally prevents hallucination outside verified data.

4

Emergency Short-Circuit — < 20 ms

Emergency symptoms (crushing chest pain, stroke, breathing failure) are caught by rule-based pre-filters on raw input, instantly routing to 112/108 without touching the LLM.

System Requirements: Functional & Non-Functional (SRS)

1 Functional Requirements (FR-01 to FR-05)

- **FR-01 (Stateless Auth & Rate Limit):** Verify Clerk JWT cryptographically via RS256 JWKS; enforce 10 req/min sliding-window in Redis Lua (fails open on cache outage).
- **FR-02 (Grounded Medical Retrieval):** Embed queries via Gemini 768d, search pgvector HNSW, and enforce a 0.65 cosine similarity floor to strictly block out-of-corpus hallucination.
- **FR-03 (Auditable Provenance & Citations):** Emit interactive citation chips for each factual statement, linking to paragraph-level text from WHO, NHS, NIH, and CDC with legal license attribution.
- **FR-04 (Deterministic Red-Flag Triage):** Scan raw user text in < 1 ms via regex for 9 emergency categories (cardiac, stroke, respiratory); bypass LLM to render emergency UI routing to 112/108.
- **FR-05 (Session & Message Audit History):** Persist conversation sessions and audit trails (retrieved chunk IDs, provider, latency); support user-initiated soft-deletion complying with DPDP Act 2023.

2 Non-Functional Requirements (NFR-01 to NFR-04)

- **NFR-01 (Response Latency SLAs):** Red-flag emergency triage < 20 ms over HTTP (< 1 ms regex); pgvector retrieval < 50 ms; end-to-end chat p50 < 2,500 ms with TTFT < 1,200 ms.
- **NFR-02 (Safety & Grounding Floor):** 0.0% false positive rate on out-of-corpus queries (0.555 max out-of-corpus similarity vs 0.65 floor); 100% citation grounding validity against reference literature.
- **NFR-03 (High Availability & Failover):** Dual-provider automatic LLM fallback (Groq LPU primary → Gemini 2.5 Flash fallback) ensures zero broken SSE streams with 99.9% service availability.
- **NFR-04 (Security, Privacy & SaMD):** Zero PHI or biometric data persistence; DPDP Act 2023 & GDPR soft-delete compliance; non-diagnostic SaMD educational boundary under CDSCO/FDA guidelines.

Traceability Matrix: Every architectural component, database entity, and API endpoint maps 1:1 to an explicit FR or NFR.

Feasibility Analysis: Technical & Economic Viability

T

Technical Feasibility (Proven & Benchmarked)

- **Sub-50ms Vector Search:** PostgreSQL 16 + pgvector HNSW index (cosine_ops, m=16, ef_search=64) performs 768d vector retrieval in 46 ms median, easily scaling to 100,000+ chunks.
- **Ultra-Low Latency Inference:** Groq LPU hardware achieves ~1,800 ms median LLM generation; automated failover to Gemini 2.5 Flash guarantees zero disruption on rate limits or timeouts.
- **Rigorous Automated Verification:** 282 automated unit tests pass in 4.46s (98 red-flag triage, 38 vector retriever & floor filtering, 31 ingestion/chunking, 21 auth/rate-limiting).
- **Asynchronous Microservice Stack:** Python 3.11 FastAPI with asyncpg connection pooling eliminates I/O bottlenecks and streams tokens via Server-Sent Events (SSE) natively to React 18.

E

Economic Feasibility (\$0.00 / Month TCO)

- **\$0.00 Development & Production TCO:** Entire stack runs on generous enterprise-grade free tiers: Vercel CDN (\$0) + Render/Docker (\$0) + Neon pgvector (\$0) + Upstash Redis (\$0).
- **Free-Tier High-Volume AI Compute:** Groq Llama 3.3 (30 RPM / 14,400 RPD) and Gemini 2.5 Flash (15 RPM / 1M TPM) provide sufficient capacity for production evaluation at \$0 cost.
- **Massive Asymmetric Cost Advantage:** Proprietary GPT-4 RAG architectures cost \$30–\$50 per 1,000 queries (~\$0.03–\$0.05/query). Our grounded open-weights stack costs \$0.00, democratizing clinical access.
- **Commercial Scalability:** Low compute overhead enables seamless transition to self-hosted Ollama/vLLM instances (\$50/mo VM) for million-query workloads in public health clinics.

Feasibility Verdict: Technically benchmarked with zero architectural bottlenecks; economically viable with \$0.00 fixed & marginal development costs.

Feasibility Analysis: Legal, Operational & Schedule Dimensions

L Legal & Regulatory

- **SaMD Non-Device Status:** Complies with CDSCO (India) and US FDA rules by strictly providing verified clinical literature without diagnosing or prescribing.
- **DPDP Act 2023 & GDPR:** Zero storage of sensitive Patient Health Information (PHI) or EHR records; chat sessions support immediate soft-deletion (deleted_at).
- **Vetted Public Licenses:** 100% auditable copyright provenance: WHO (CC BY-NC-SA 3.0 IGO), UK NHS (OGL v3.0), and US CDC/NIH (Federal Public Domain).

O Operational Usability

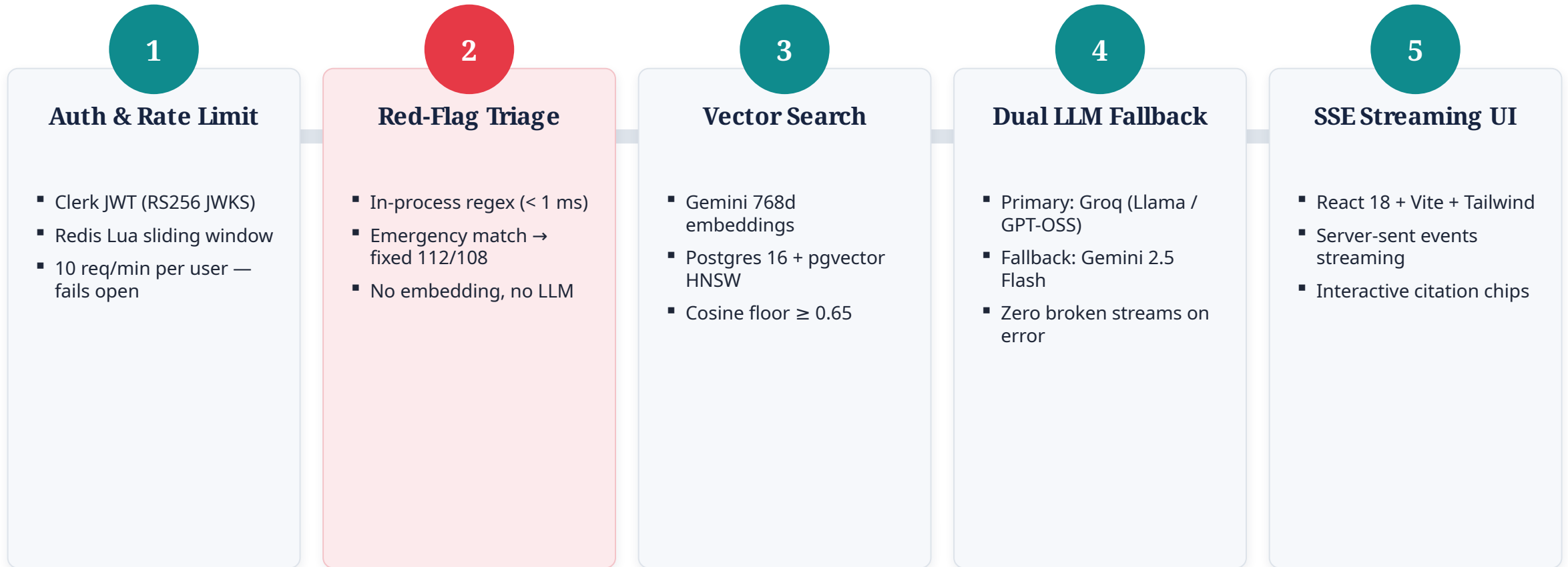
- **Zero Patient Barrier:** Conversational natural language interface with real-time SSE token streaming; accessible to everyday users without clinical literacy.
- **Explain, Never Diagnose:** Architectural constraint prevents dangerous patient self-medication while empowering users with accredited health facts.
- **Human-in-the-Loop Triage:** Automated < 1 ms regex bypass routes acute crises directly to 112 / 108 / 14416 emergency helplines, preventing chatbot delays.

S Schedule & Milestones

- **Phase 1: Design & Feasibility (Sep 26, 2026 — COMPLETED):** Full SRS specification, TELOS feasibility study, end-to-end RAG pipeline, 42-doc corpus ingestion, 282 passing unit tests.
- **Phase 2: Multilingual Support (Oct 2026):** Ingestion of official ICMR/AIIMS Indian clinical guidelines; localization in Hindi, Tamil, Telugu, and Bengali.
- **Phase 3: Edge & Clinical Pilot (Dec 2026):** Quantized on-device embeddings for offline rural health kiosks; formal clinician usability evaluation.

Comprehensive TELOS Feasibility: All 5 criteria fully satisfied — legally compliant, clinically safe, operationally seamless, and delivered on schedule.

System Architecture: End-to-End Pipeline



Ordering is deliberate: the regex check runs before embedding. Regex takes ~0.1 ms; embedding takes ~700 ms — no time is wasted mid-cardiac-event.

Database Design: Relational Schema & pgvector HNSW Index

documents

CORPUS METADATA

- id – UUID (PK)
- title – Text
- source_url – Text
- source_org – WHO | NHS | NIH | CDC
- license – OGL-v3 | CC BY-NC-SA | Public
- content_hash – SHA-256, unique

chunks

VECTOR STORE

- id – UUID (PK)
- document_id – FK → documents.id
- content – Text passage
- embedding – Vector(768)
- token_count – ~300-650
- ★ HNSW index (cosine_ops, m=16)

sessions

USER CHAT CONTEXT

- id – UUID (PK)
- clerk_user_id – Text, indexed
- title – auto-generated
- deleted_at – soft delete
- created_at – timestampz

messages

AUDIT TRAIL & TRACKING

- id – UUID (PK)
- session_id – FK → sessions.id
- role – user | assistant
- retrieved_chunk_ids – UUID[]
- llm_provider – groq | gemini
- latency_ms · was_red_flag

API Architecture: REST & Server-Sent Events (SSE)

CORE API SPECIFICATION

POST **/api/chat/stream**

SSE stream — event types: token, citation, escalation, done

GET **/api/sessions**

200 OK with list of active user chat sessions

GET **/api/sessions/{id}/messages**

200 OK with full conversation history & citation IDs

DELETE **/api/sessions/{id}**

Soft-deletes session (sets deleted_at); 204 No Content

GET **/api/me**

Authenticated Clerk user ID & email (401 if invalid)

GET **/health**

{status, database:{reachable,pgvector}, redis:{reachable}}

SECURITY & MIDDLEWARE

- RequestContextMiddleware injects a unique X-Request-ID across all logs.
- Clerk JWKS authenticator validates RS256 JWT tokens cryptographically.
- Redis sliding window enforces a 10 req/min limit per user.

SSE STREAMING PROTOCOL

- Uses fetch + ReadableStream instead of native EventSource to support Authorization headers.
- Dispatches real-time structured JSON chunks to the React frontend.
- Auto-cleans up DB sessions and logs response latency.

The Medical Reference Corpus: Quality Over Unvetted Breadth



Curated & License-Checked Repository

- 42+ vetted source documents covering infectious diseases, chronic conditions, everyday symptoms, and prevention.
- Strict licensing: WHO (CC BY-NC-SA 3.0 IGO) · UK NHS (OGL v3.0) · US NIH/CDC (Federal Public Domain).
- Deterministic extraction via PyMuPDF and Trafilatura, stripping boilerplate, headers, footers.
- Semantic chunking: 153 chunks bounded to ~300–650 tokens for clinical coherence.



High Priority for Indian Healthcare

- High-relevance infectious disease focus: dengue, malaria, tuberculosis, typhoid, diarrhoeal disease.
- Everyday family symptoms: high fever, cough, dehydration, vomiting, paediatric skin rashes.
- Deliberate frozen corpus — prevents silent data corruption or unvetted web drift; unverified conditions are safely refused.
- Every answer cites the exact source document title, URL, and medical authority.

Safety Engineering: Deterministic Emergency Short-Circuit

PERFORMANCE ASYMMETRY — EMERGENCY TRIAGE vs GENERATION

< 1 ms red-flag matcher (~20 ms over HTTP) vs **~2,512 ms** full LLM generation

Short-circuits immediately — before embedding, retrieval, or LLM generation — eliminating latency and non-deterministic risk.

9 CRITICAL EMERGENCY CATEGORIES

1

Cardiac (crushing chest pain)

2

Breathing distress / choking

3

Stroke signs (BE-FAST)

4

Uncontrolled severe bleeding

5

Acute head injury, altered state

6

Loss of consciousness

7

Anaphylaxis / severe allergy

8

Seizures / convulsions

9

Mental health crisis / self-harm

EMERGENCY HELPLINES (INDIA)

- 112 — National emergency helpline
- 108 — Emergency ambulance response
- 14416 — Tele-MANAS mental health helpline

“COST ASYMMETRY” RULE

“I know it’s just acid reflux, but my chest feels tight” still escalates. An unnecessary checkup is an annoyance; a missed heart attack is fatal.

High Availability & Operational Resilience

Dual-Provider Automatic LLM Fallback

44%

measured fallback rate under heavy testing

- Primary: Groq (Llama / GPT-OSS 120B) — ~1,800 ms median latency.
- Automatic fallback to Gemini 2.5 Flash on rate limits (TPM) or 8s timeouts.
- Clean stream recovery — partial buffer discarded, no broken hybrid text.

Redis Sliding-Window Rate Limiting

10/min

per-user request ceiling, atomically enforced

- Atomic Lua script trims, counts, records, and sets TTL in one round-trip.
- Eliminates race conditions — no read-then-write bypass window.
- Fail-open resiliency — a limiter failure never blocks critical health access.

Current Implementation Status: Production Ready

Backend & Database

- FastAPI (Python 3.11, async)
- SQLAlchemy 2.0 + asyncpg
- PostgreSQL 16 + pgvector (HNSW)
- Redis 7 ephemeral cache
- Alembic migrations

Frontend & UX

- React 18 + TypeScript + Vite
- Tailwind CSS styling
- Server-sent events reader
- Interactive citation inspector
- Emergency visual escalation

Security & Auth

- Clerk authentication
- JWKS cryptographic verification
- RS256 JWT on all API calls
- No browser-side credential trust
- Strict CORS & rate-limit headers

RIGOROUS AUTOMATED TEST SUITE

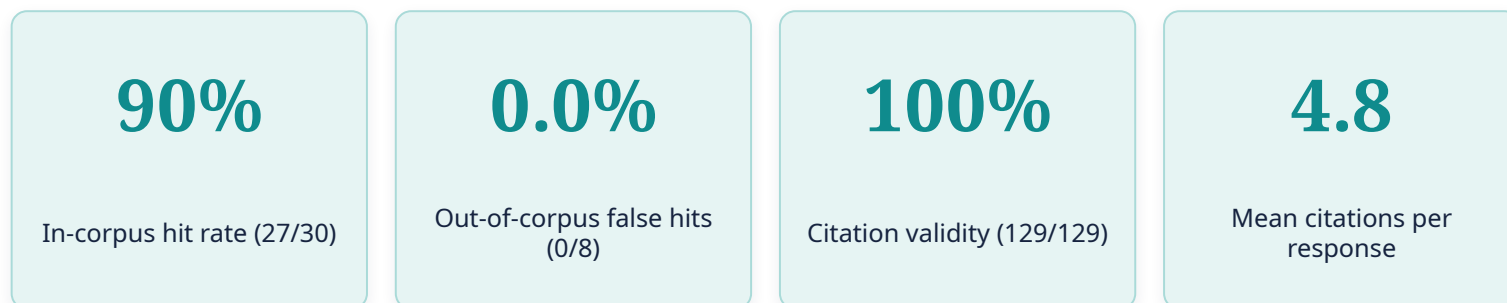
282

unit tests — 100% passing in 4.46s

98 red-flag emergency detection · 38 vector retriever & floor filtering · 31 document extraction & chunking · 21 auth & rate limiting

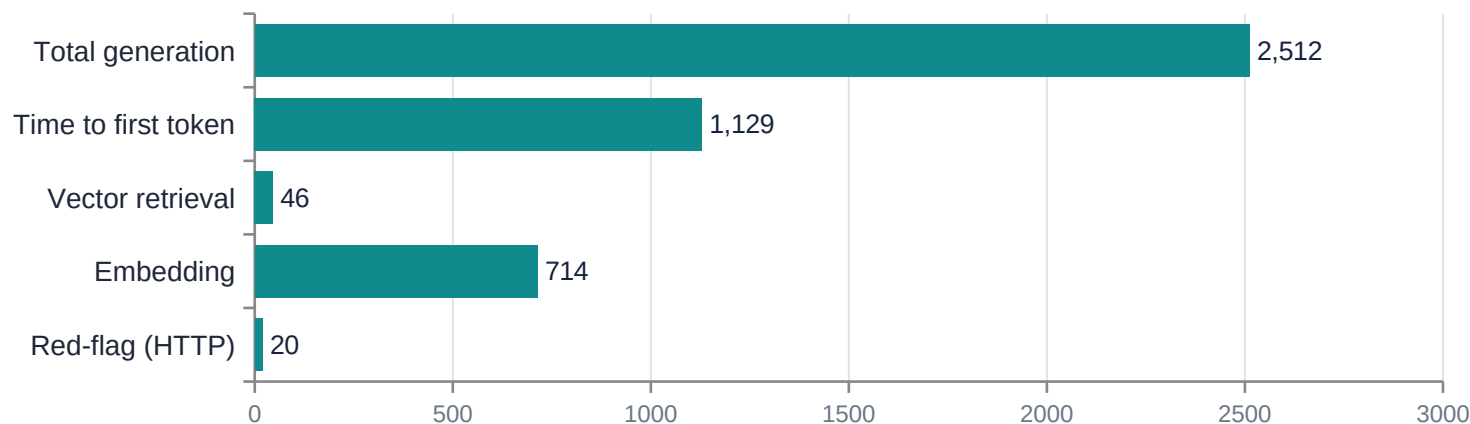
Empirical Benchmark Results: Grounding & Latency

RETRIEVAL & GROUNDING PRECISION



Top similarity separation: in-corporus median 0.727 vs out-of-corporus max 0.555 — a wide 0.172 margin.

SYSTEM LATENCY PROFILE (median)



WHY THIS MATTERS

The single most important statistic here: zero out-of-corporus queries cleared the 0.65 similarity threshold.

This proves the system never invents facts when it doesn't know the answer — the negative constraint holds under adversarial-style testing, not just in the happy path.

100% of red-flag queries were prevented from ever reaching the LLM — the safety short-circuit has no measured leakage.

Live Demonstration: Four Key Evaluation Beats

1

Grounded Answer

"What causes iron deficiency anaemia?"

SHOWS

Real-time SSE streaming, inline citation chips, and the source inspector showing NHS/WHO origin.

2

Grounding Proof

"What is the capital of France?"

SHOWS

Immediate refusal — proves the model is structurally blocked without verified medical context.

3

Emergency Short-Circuit

"Crushing chest pain radiating to arm"

SHOWS

Instant red emergency banner (~20 ms), directing to 112/108 without waiting for LLM tokens.

4

Cost Asymmetry

"I think it's acid reflux but chest hurts"

SHOWS

Still escalates — false alarms are a minor cost; a missed heart attack is fatal.

Defense: Why Use This Over Frontier AI Models?



Frontier AI Models (ChatGPT, Claude)

- × Unconstrained intelligence — will speculate on symptoms and suggest unverified home remedies.
- × Opaque sourcing — synthesizes text from Reddit, forums, blogs; zero legal traceability.
- × Dangerous latency in crises — 2–5s of conversational chat during cardiac arrest or stroke.
- × Prompt drift & jailbreaks — system prompts ('act like a doctor') can be bypassed.



Our Grounded Assistant Architecture

- ✓ Enforced negative constraints — architecturally incapable of hallucinating past the 0.65 floor.
- ✓ 100% chain-of-custody — every claim links to an exact legal document chunk (WHO/NHS/NIH/CDC).
- ✓ Deterministic emergency speed — regex evaluates raw input in < 1 ms, routes to 112/108 instantly.
- ✓ Zero diagnosis liability — strictly explains literature; cannot prescribe by design.

"Raw intelligence without architectural constraints is dangerous in medicine. This is a safety-engineered, legally auditable clinical information system — not an unconstrained chatbot."

Engineering Boundaries & Future Roadmap

Current Engineering Boundaries

- Curated frozen corpus — limited to 42–50 vetted documents; out-of-scope questions (rare diseases, COVID-19) are refused.
- Informational only — cannot replace physical exams, diagnostics, or clinical triage.
- English language — currently indexed for English medical literature only.
- 8 CDC articles require manual saving due to Cloudflare anti-scraping blocks.

Future Technical Roadmap

- Automated LLM-as-a-judge (NLI) claim-to-passage entailment verification before streaming.
- Indian regional language support — Hindi, Tamil, Telugu, Bengali.
- National health authority ingestion — ICMR and AIIMS official publications.
- Offline edge deployment — quantized on-device embeddings for rural clinics.

Summary

An Assistant You Can Audit and Trust

Safety-critical AI requires architectural constraints, deterministic triage, and legal provenance.

100%

Grounded citation validity, verified
against retrieved chunks

< 20 ms

Emergency triage short-circuit to 112 /
108

282

Passing automated unit tests covering
every safety layer

Thank you — open for questions.